

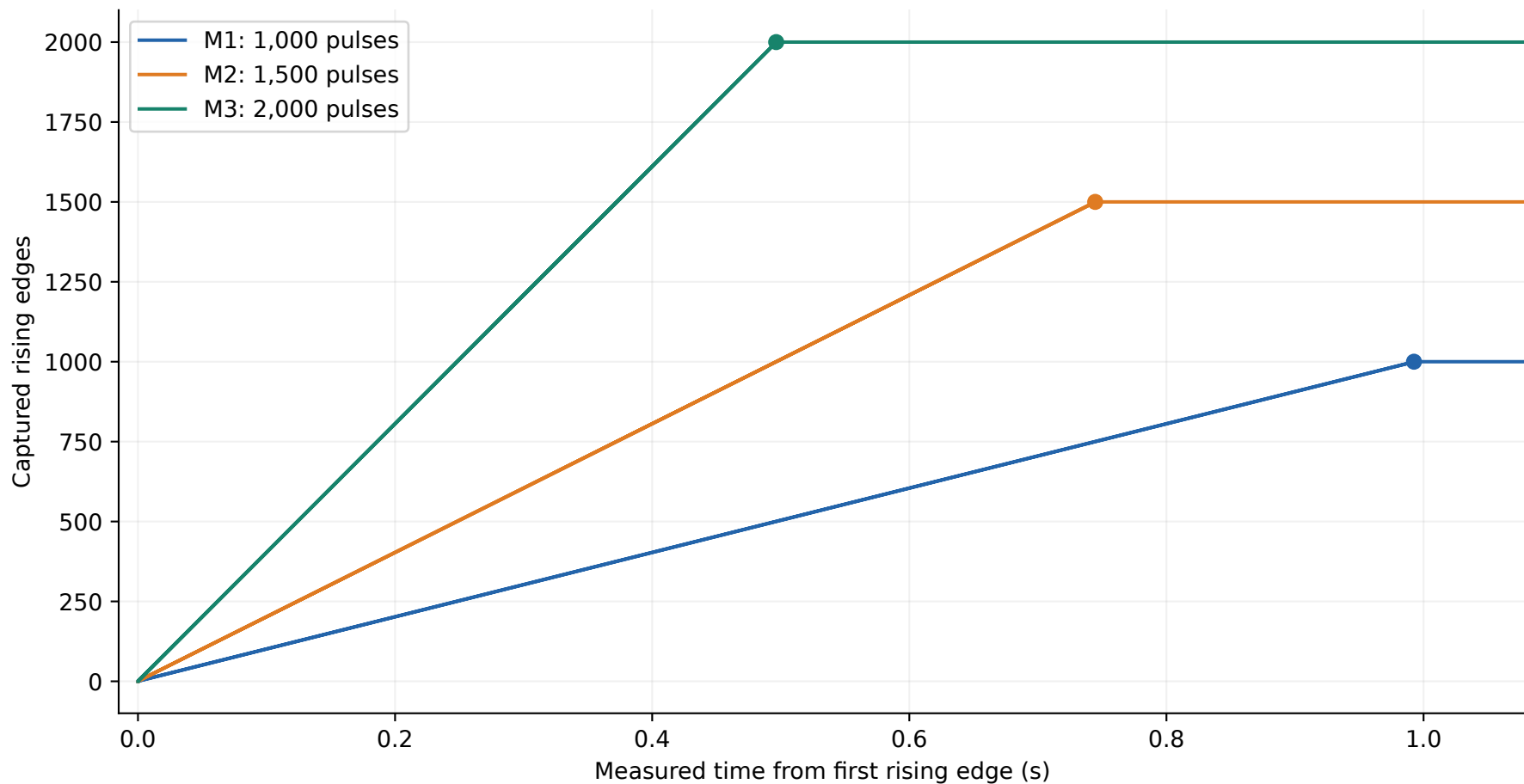
STEP hardware validation — pulse counts and stopping

Current firmware: reported pulse counts match the independent logic-analyzer captures.

Absolute clock timing remains unresolved.

Case	Captured M1/M2/M3	Reported M1/M2/M3	State	Counts
1	1/2/3	1/2/3	Completed	MATCH
2	1000/1500/2000	1000/1500/2000	Completed	MATCH
3	100000/100000/100000	100000/100000/100000	Completed	MATCH
4	1024/1024/1024	1024/1024/1024	Completed	MATCH
5	995/498/249	995/498/249	Stopped	MATCH
6	1000/1000/1000	1000/1000/1000	Completed	MATCH
7	512/256/128	512/256/128	Underrun	MATCH
8	435/435/435	435/435/435	Completed	MATCH

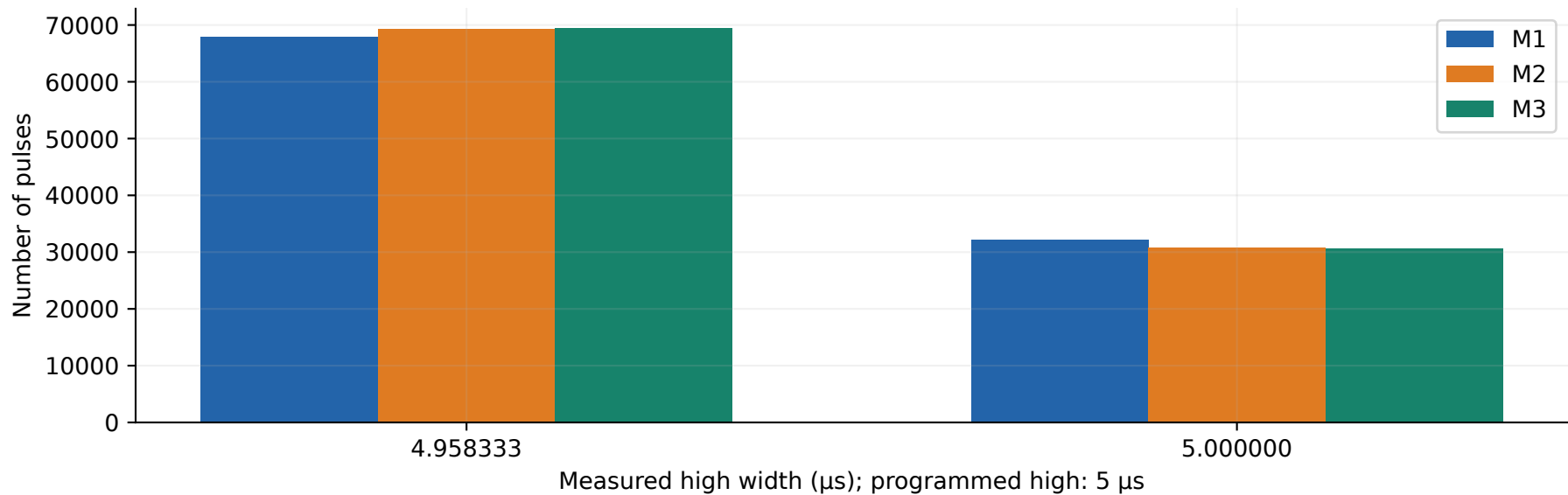
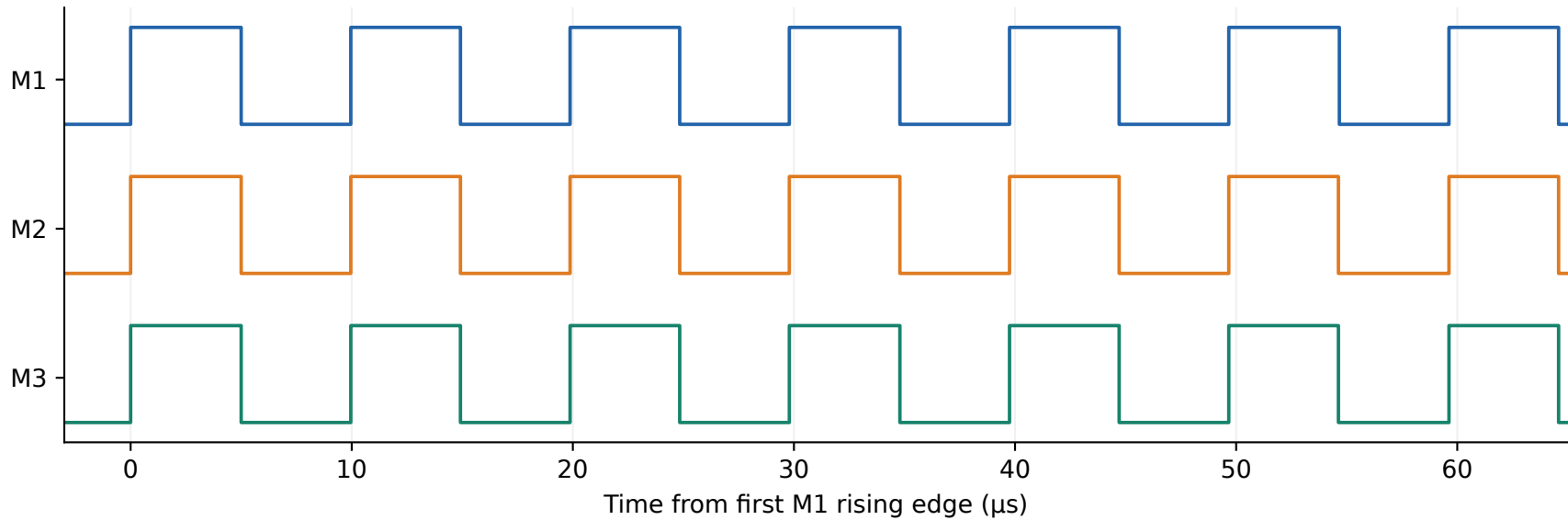
Case 2 — independent rates and finish times



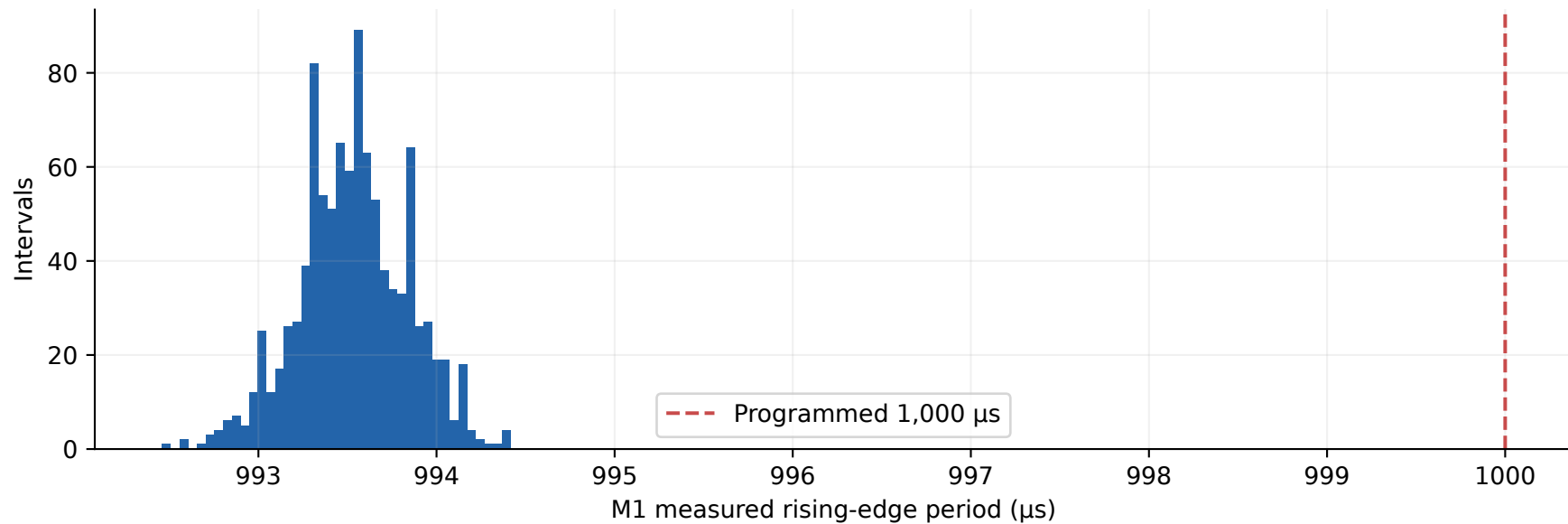
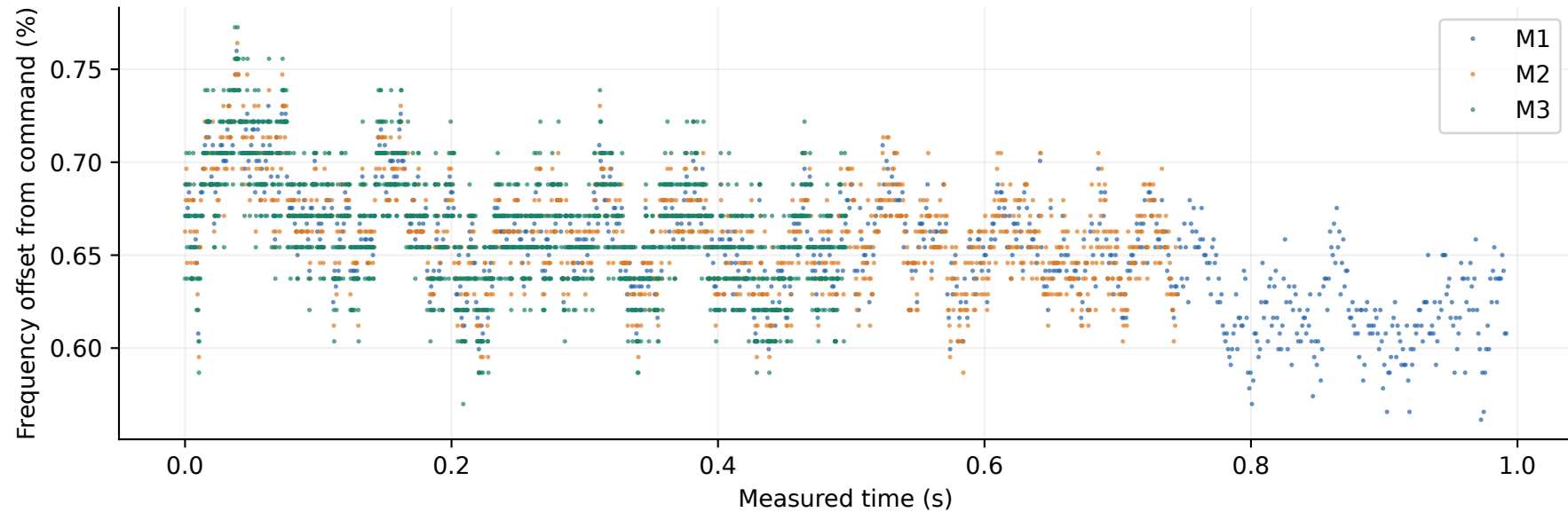
All first edges align within one 41.7 ns sample. Dots mark the final falling edges; no subsequent pulses in the capture.

Case 3 — 100,000 pulses/axis; measured M1 rate 100.643 kHz

First pulses: all three axes share the first edge

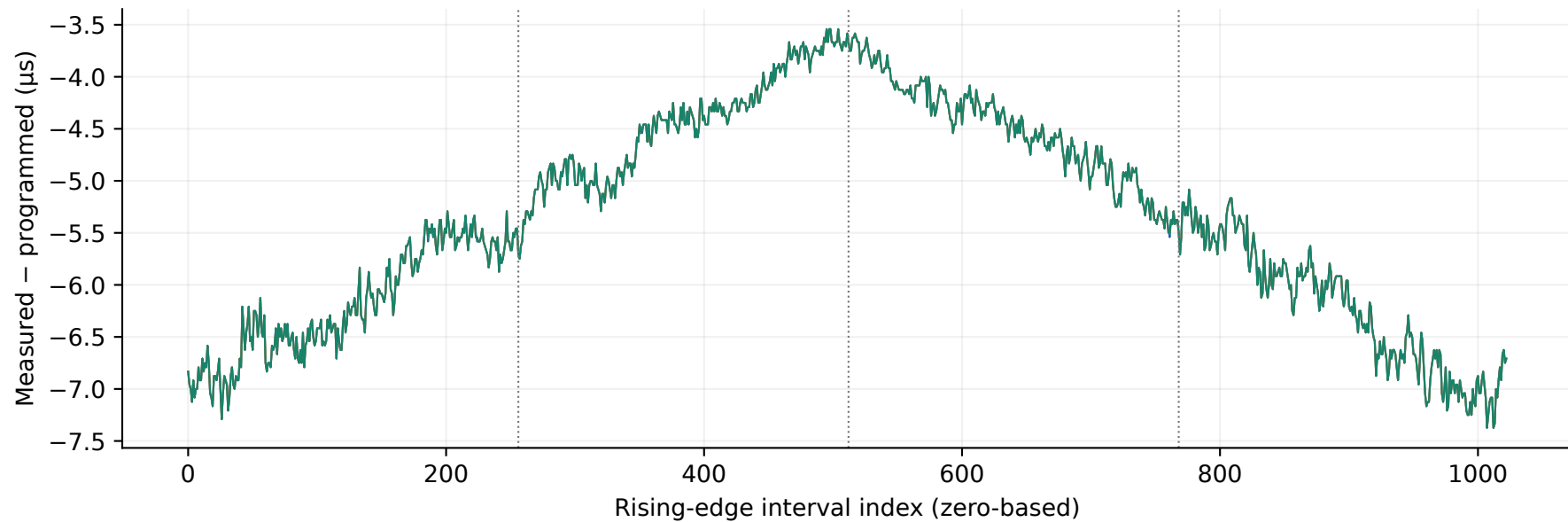
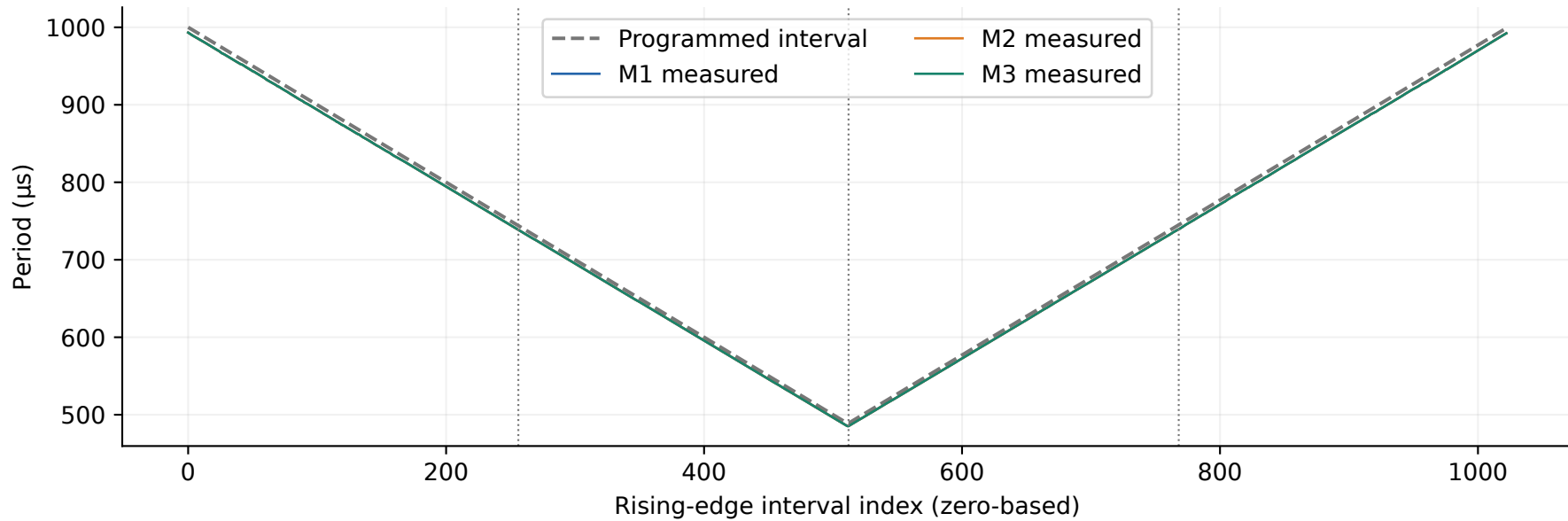


Clock accuracy finding — frequency offset and variation remain



Timing is relative to the analyzer clock. Independent Hantek measurements also show a fast STEP rate; absolute calibration remains open.

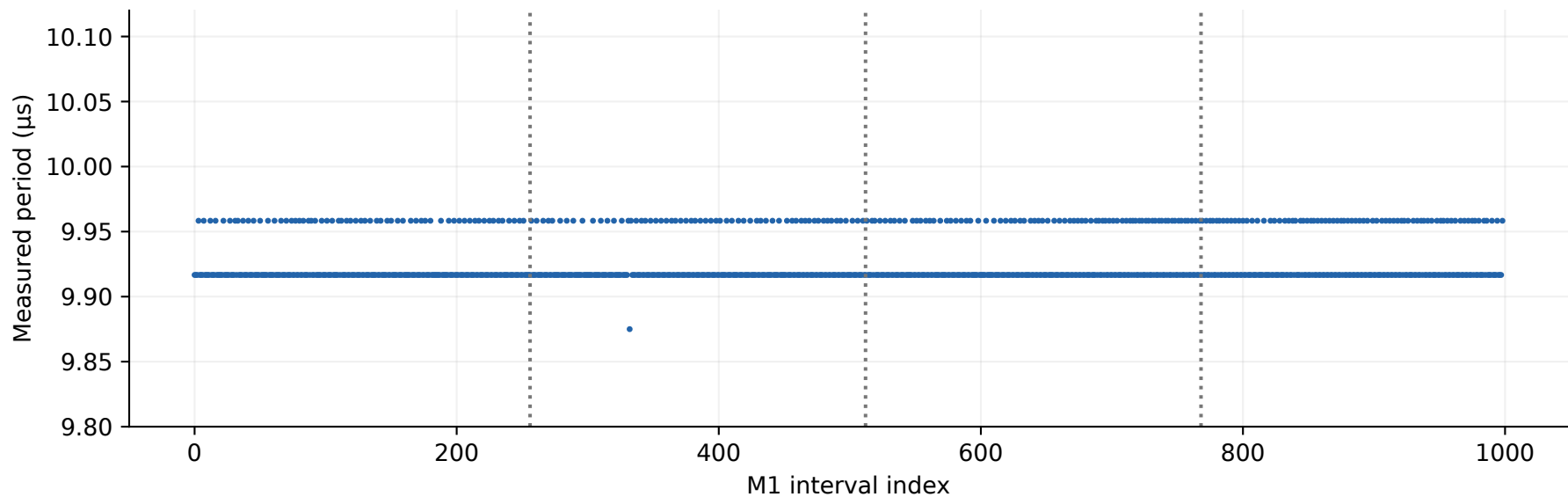
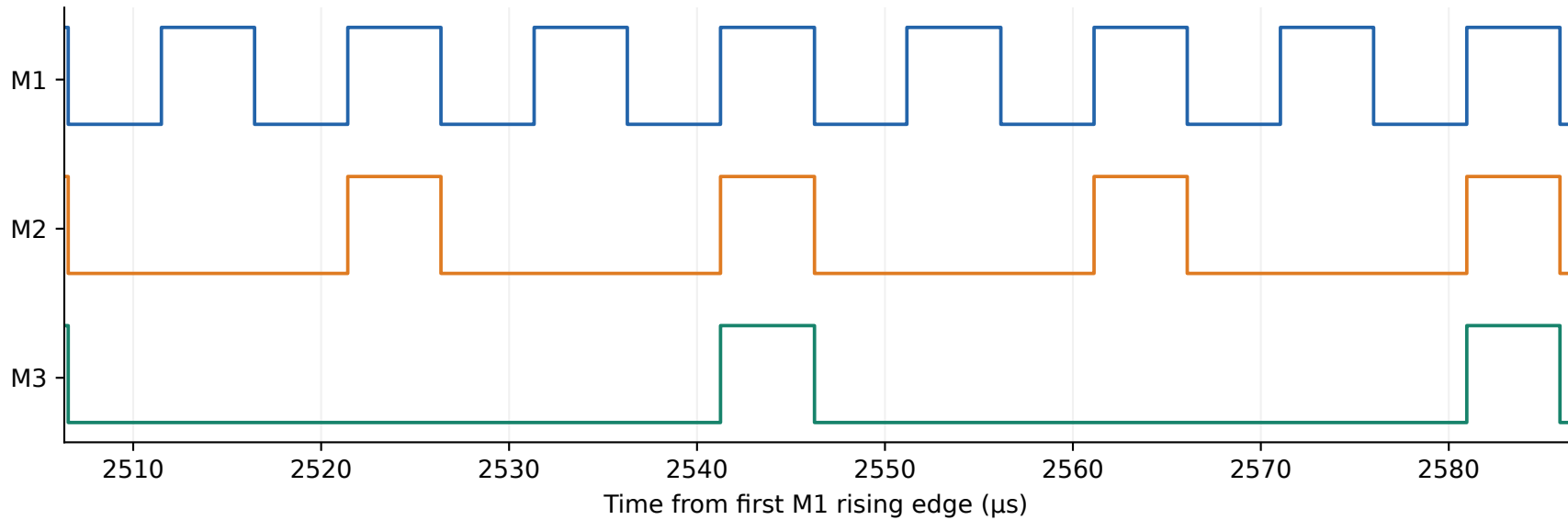
Case 4 — copied acceleration/deceleration profile, 1,024 pulses/axis



Dotted lines indicate 256-pulse buffer boundaries. All counts match; systematic timing error remains visible.

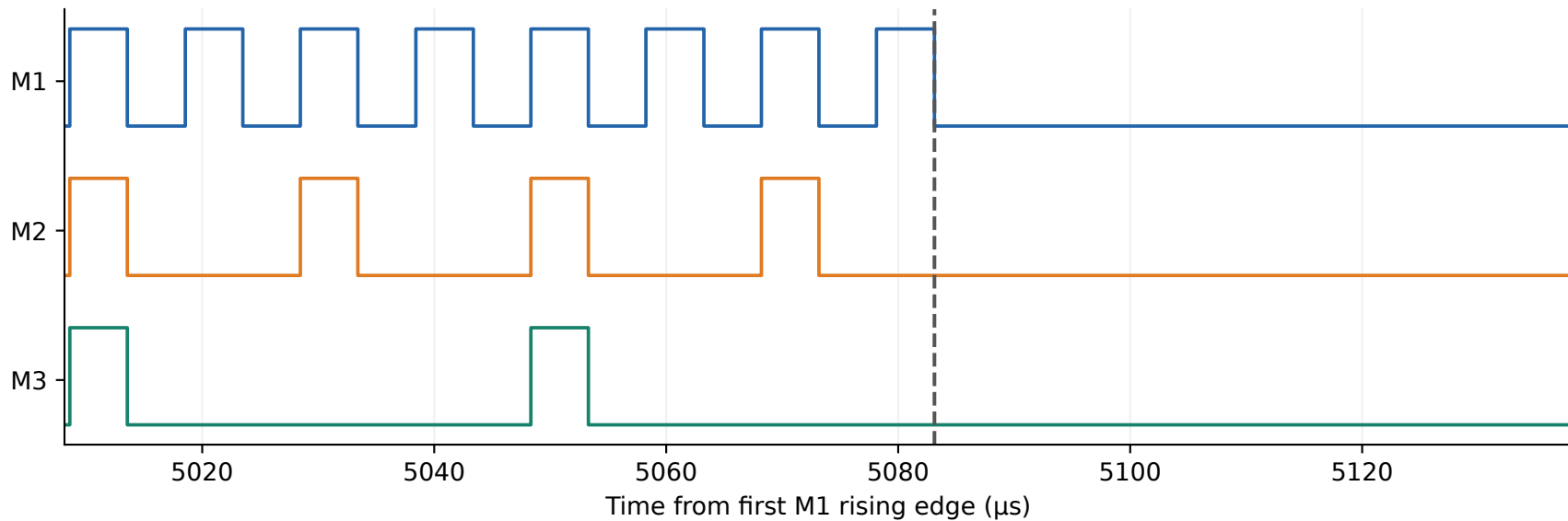
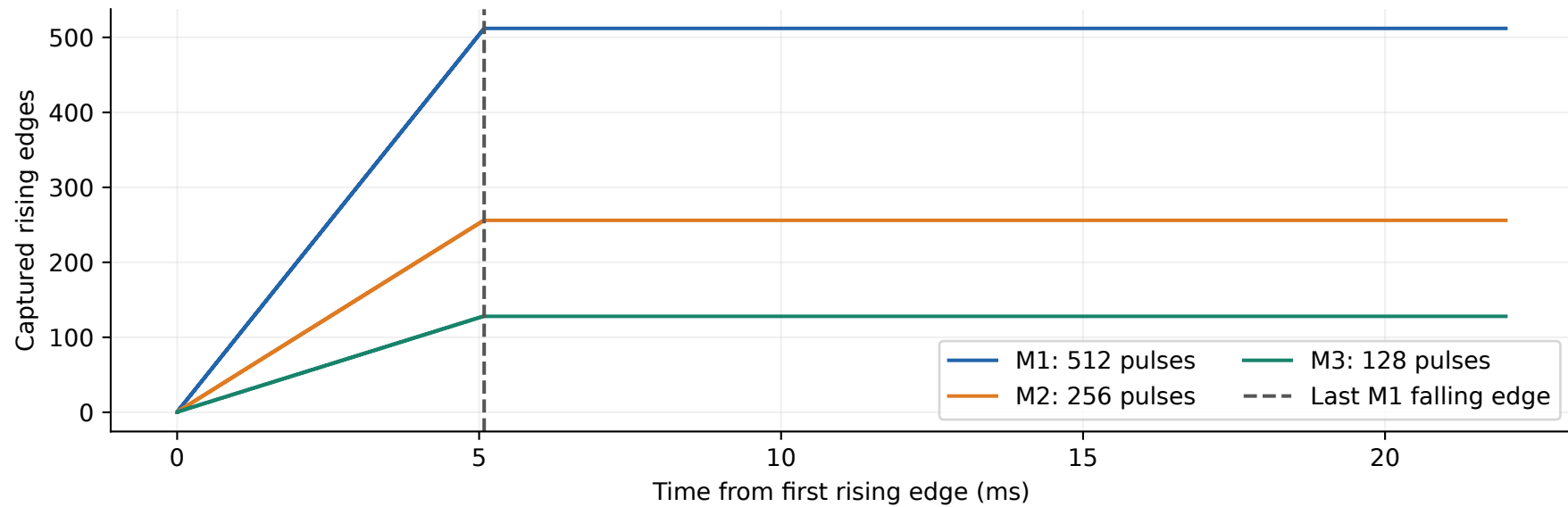
Case 6 — delayed refill IRQ: 1,000 captured pulses on every axis

Pulse train crosses the first M1 DMA-buffer boundary while its IRQ is withheld



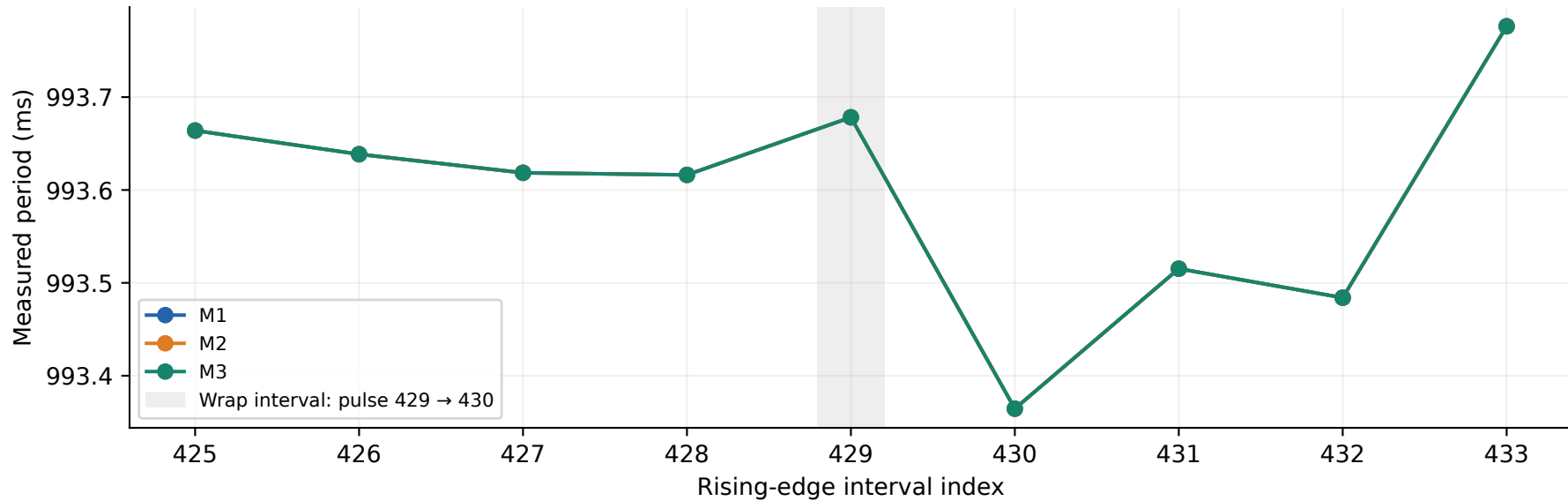
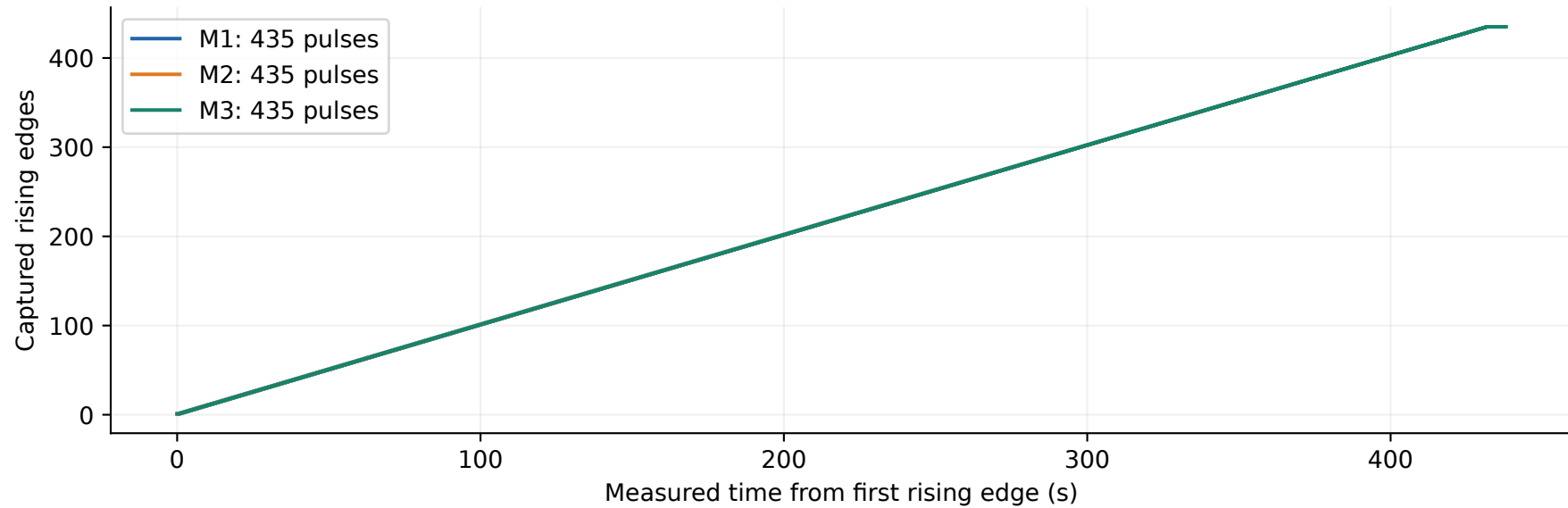
The firmware masks DMA IRQs for about 4.5 ms including the start delay. The analyzer observes the outputs, not the IRQ mask directly.

Case 7 — autonomous stop with all DMA IRQs withheld



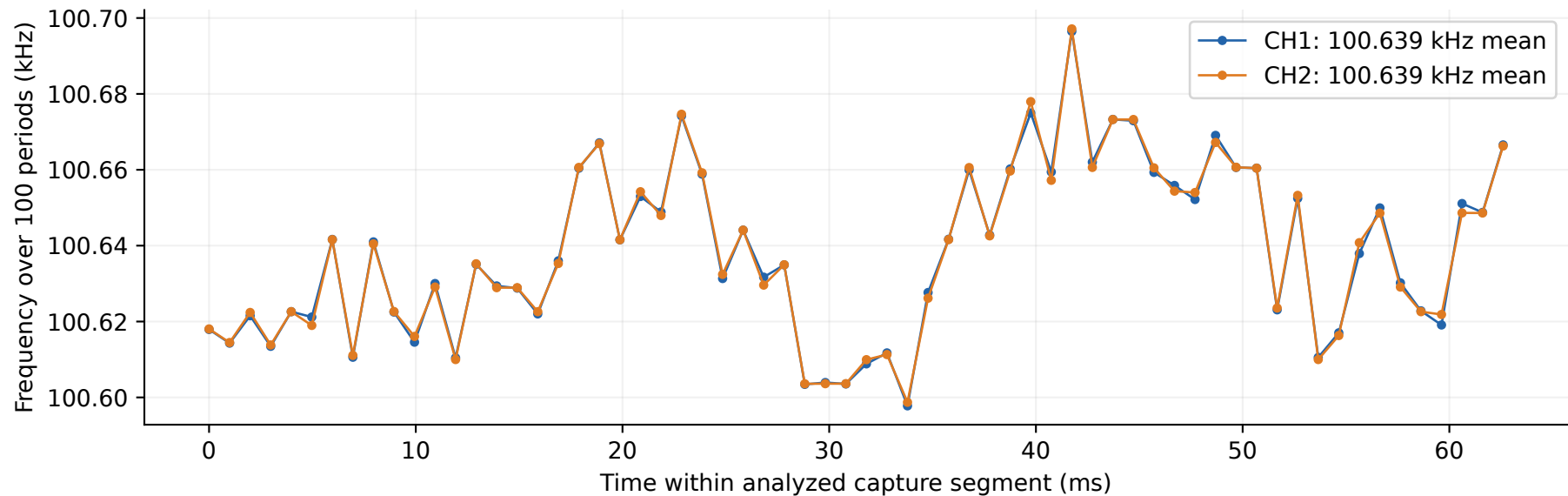
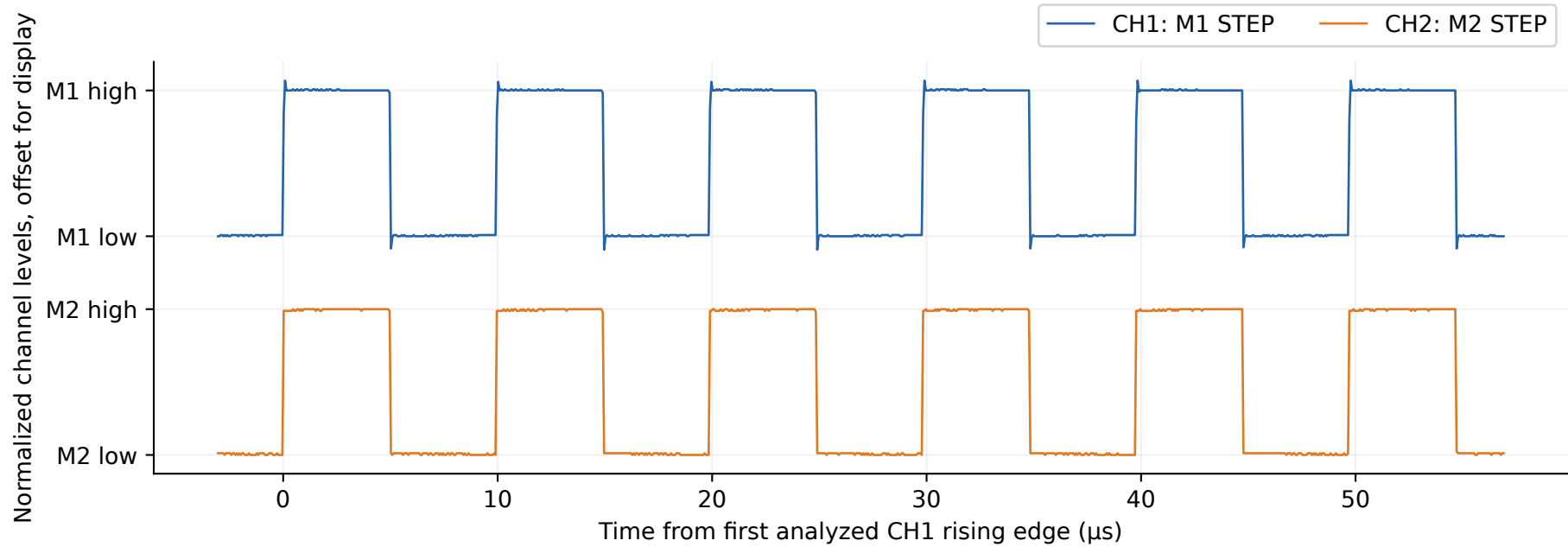
UART: Underrun, exact=1, counts=512/256/128, guard_stopped=1 before IRQ restoration. Outputs remain low after the final pulses.

Case 8 — real 32-bit timer rollover, 435 pulses on every axis

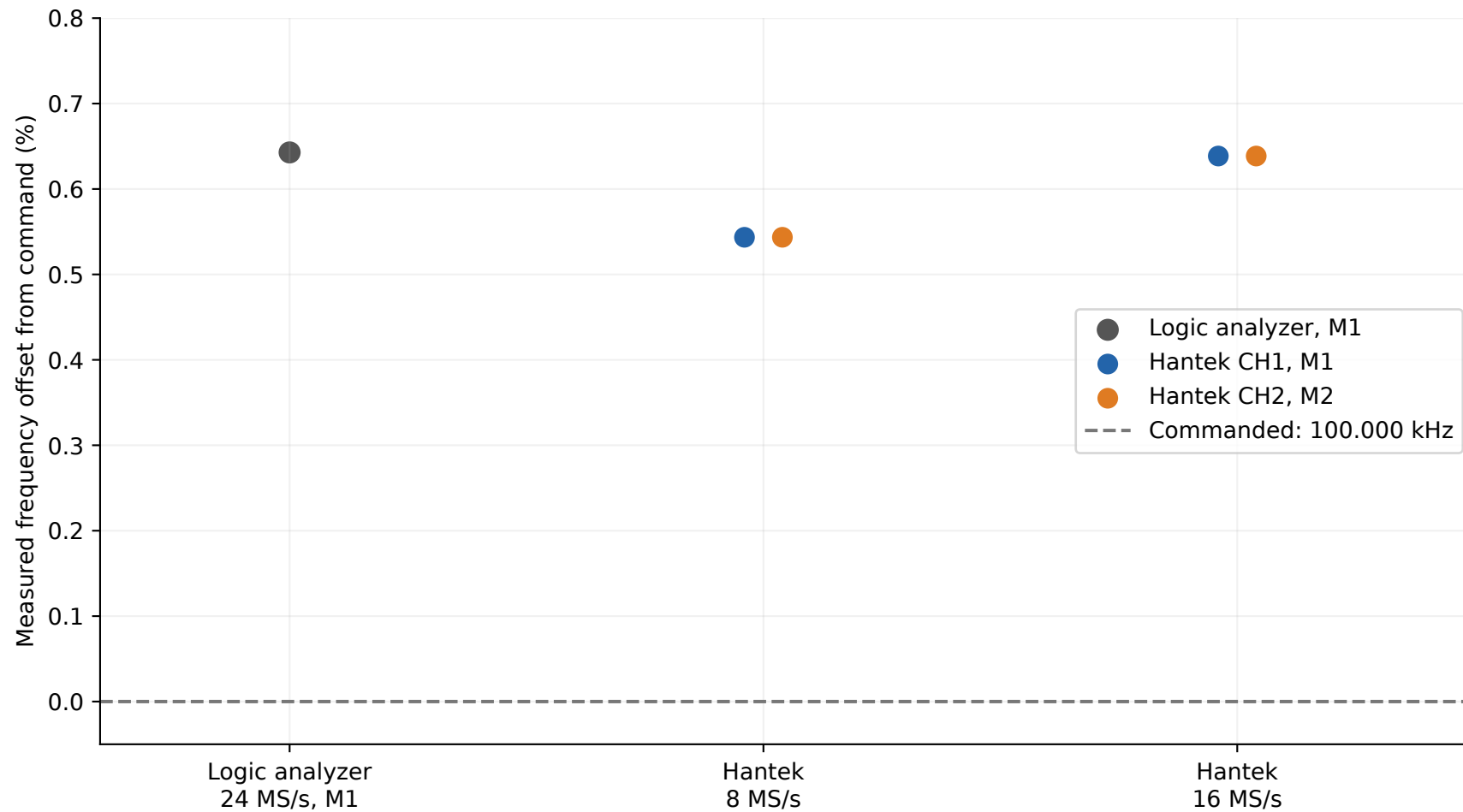


4 MS/s capture; 250 ns sample spacing resolves the 5 μ s highs. The rollover interval is located from scheduled timer ticks; no forced wrap.

Hantek 6022BE — simultaneous M1 and M2 STEP measurements

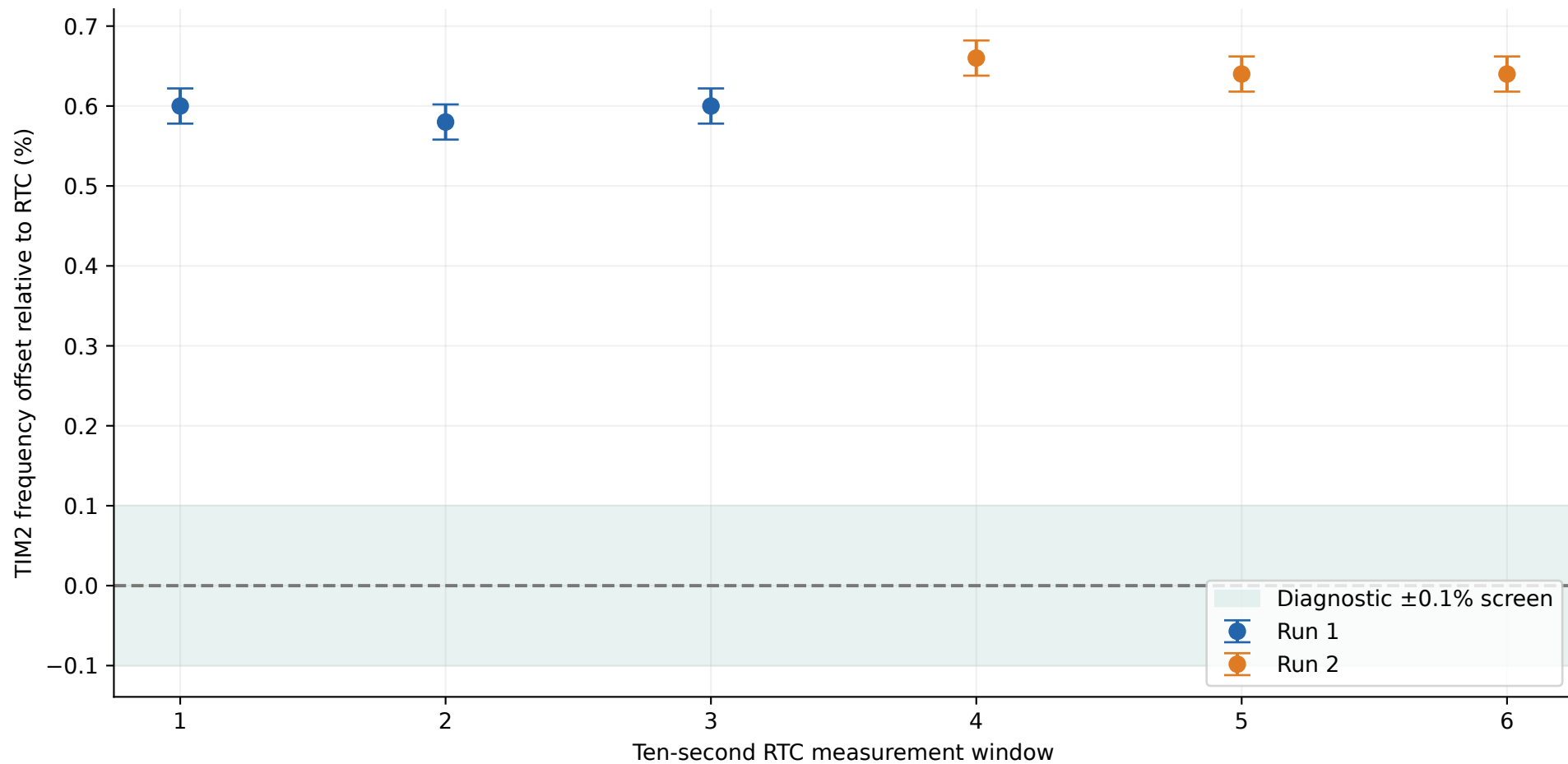


Independent instruments both measure a fast STEP rate



Same firmware, separate bursts. Channel pairs are simultaneous within each Hantek capture. No instrument has an independent timebase calibration.

TIM2 measured against the independent RTC crystal



No external instrument. STEP held low. Error bars cover polling and synchronization; LSE crystal tolerance is additional.